

1) Start Hdfs

```
nitish@nitish-L00-15IRX9:~$ start-dfs.sh
Starting namenodes on [localhost]
localhost: namenode is running as process 32684. Stop it first and ensure /tmp/hadoop-nitish-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 32875. Stop it first and ensure /tmp/hadoop-nitish-datanode.pid file is empty before retry.
Starting secondary namenodes [nitish-L00-15IRX9]
nitish-L00-15IRX9: secondarynamenode is running as process 33132. Stop it first and ensure /tmp/hadoop-nitish-secondarynamenode.pid file is empty before retry.
nitish@nitish-L00-15IRX9:~$ start-yarn.sh
Starting resourcemanager
resourcemanager is running as process 34737. Stop it first and ensure /tmp/hadoop-nitish-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 34899. Stop it first and ensure /tmp/hadoop-nitish-nodemanager.pid file is empty before retry.
nitish@nitish-L00-15IRX9:~$
```

2)

```
nitish@nitish-L00-15IRX9:~$ jps
34737 ResourceManager
34899 NodeManager
38297 Jps
32875 DataNode
32684 NameNode
33132 SecondaryNameNode
nitish@nitish-L00-15IRX9:~$
```

3)

```
nitish@nitish-L00-15IRX9:~$ jar -xvf $HADOOP_HOME/share/hadoop/mapreduce/sources/hadoop-mapreduce-examples-3.3.6-sources.jar
created: META-INF/
```

4) Navigate to path for WordCount.java

```

inflatel: org/apache/hadoop/examples/terasort/Random101.java
inflatel: org/apache/hadoop/examples/terasort/TeraGen.java
inflatel: org/apache/hadoop/examples/terasort/TeraChecksum.java
inflatel: org/apache/hadoop/examples/terasort/TeraScheduler.java
inflatel: org/apache/hadoop/examples/terasort/package.html
inflatel: org/apache/hadoop/examples/terasort/TeraSort.java
inflatel: org/apache/hadoop/examples/terasort/2009-write-up/1PBTaskTime.png
inflatel: org/apache/hadoop/examples/terasort/2009-write-up/500GBTaskTime.png
inflatel: org/apache/hadoop/examples/terasort/2009-write-up/tera.bib
inflatel: org/apache/hadoop/examples/terasort/2009-write-up/1TBTaskTime.png
inflatel: org/apache/hadoop/examples/terasort/2009-write-up/Yahoo2009.tex
inflatel: org/apache/hadoop/examples/terasort/2009-write-up/100TBTaskTime.png
inflatel: org/apache/hadoop/examples/terasort/TeraOutputFormat.java
inflatel: org/apache/hadoop/examples/QuasiMonteCarlo.java
inflatel: org/apache/hadoop/examples/WordMedian.java
inflatel: org/apache/hadoop/examples/AggregateWordHistogram.java
inflatel: org/apache/hadoop/examples/ExampleDriver.java
inflatel: org/apache/hadoop/examples/pi/Util.java
inflatel: org/apache/hadoop/examples/pi/TaskResult.java
inflatel: org/apache/hadoop/examples/pi/DistSum.java
inflatel: org/apache/hadoop/examples/pi/math/ArithmeticProgression.java
inflatel: org/apache/hadoop/examples/pi/math/LongLong.java
inflatel: org/apache/hadoop/examples/pi/math/Montgomery.java
inflatel: org/apache/hadoop/examples/pi/math/Modular.java
inflatel: org/apache/hadoop/examples/pi/math/Bellard.java
inflatel: org/apache/hadoop/examples/pi/math/package.html
inflatel: org/apache/hadoop/examples/pi/math/Summation.java
inflatel: org/apache/hadoop/examples/pi/SummationWritable.java
inflatel: org/apache/hadoop/examples/pi/Parser.java
inflatel: org/apache/hadoop/examples/pi/package.html
inflatel: org/apache/hadoop/examples/pi/Container.java
inflatel: org/apache/hadoop/examples/pi/DistBbp.java
inflatel: org/apache/hadoop/examples/pi/Combinable.java
inflatel: org/apache/hadoop/examples/Grep.java
inflatel: org/apache/hadoop/examples/SecondarySort.java
inflatel: org/apache/hadoop/examples/WordCount.java
inflatel: org/apache/hadoop/examples/Join.java
inflatel: org/apache/hadoop/examples/DBCountPageView.java
inflatel: org/apache/hadoop/examples/MultiFileWordCount.java
inflatel: org/apache/hadoop/examples/package.html
inflatel: org/apache/hadoop/examples/WordStandardDeviation.java
inflatel: org/apache/hadoop/examples/WordMean.java
inflatel: org/apache/hadoop/examples/AggregateWordCount.java
inflatel: org/apache/hadoop/examples/Sort.java
nitish@nitish-L0Q-15IRX9:~$

```

```

nitish@nitish-L0Q-15IRX9:~$ cd org/apache/hadoop/examples
nitish@nitish-L0Q-15IRX9:~/org/apache/hadoop/examples$ ls
AggregateWordCount.java  dancing  Grep.java  package.html  RandomTextWriter.java  Sort.java  WordMean.java
AggregateWordHistogram.java  DBCountPageView.java  Join.java  pi  RandomWriter.java  terasort  WordMedian.java
BaileyBorweinPlouffe.java  ExampleDriver.java  MultiFileWordCount.java  QuasiMonteCarlo.java  SecondarySort.java  WordCount.java  WordStandardDeviation.java
nitish@nitish-L0Q-15IRX9:~/org/apache/hadoop/examples$ cat WordCount.java
/**
 * WordCount.java
 *
 * Copyright 2009 The Apache Software Foundation
 *
 * Licensed under the Apache License, Version 2.0 (the "License");
 * you may not use this file except in compliance with the License.
 * You may obtain a copy of the License at
 *
 * http://www.apache.org/licenses/LICENSE-2.0
 *
 * Unless required by applicable law or agreed to in writing, software
 * distributed under the License is distributed on an "AS IS" BASIS,
 * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
 * See the License for the specific language governing permissions and
 * limitations under the License.
 */

```

5) Copy the code and got to root directory by entering cd

Note : Dont Copy the the line package org.apache.hadoop.examples;

6) create your directory and nano data.txt file (Write data for input to wordcount)

Create WordCount.java and paste the code

7)

```

nitish@nitish-L0Q-15IRX9:~$ mkdir 31359
nitish@nitish-L0Q-15IRX9:~$ cd 31359
nitish@nitish-L0Q-15IRX9:~/31359$ nano data.txt
nitish@nitish-L0Q-15IRX9:~/31359$ nano WordCount.java

```

8)

```
nitish@nitish-L0Q-15IRX9:~/31359$ javac -classpath 'hadoop classpath' WordCount.java
nitish@nitish-L0Q-15IRX9:~/31359$ jar -cf WordCount.jar WordCount*.class
```

9)

```
nitish@nitish-L0Q-15IRX9:~/31359$ hadoop jar WordCount.jar WordCount /31359/input /31359/output
2026-05-05 18:13:31,476 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
```

10)

```
nitish@nitish-L0Q-15IRX9:~/31359$ hdfs dfs -mkdir /31359
nitish@nitish-L0Q-15IRX9:~/31359$ hdfs dfs -mkdir /31359/input
nitish@nitish-L0Q-15IRX9:~/31359$ hdfs dfs -put data.txt /31359/input
nitish@nitish-L0Q-15IRX9:~/31359$ hdfs dfs -ls /31359/input
Found 1 items
-rw-r--r--  1 nitish supergroup      120 2026-05-05 18:09 /31359/input/data.txt
```

11)

```
nitish@nitish-L0Q-15IRX9:~/31359$ hadoop jar WordCount.jar WordCount /31359/input /31359/output
2026-05-05 18:13:31,476 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
```

12)

```
nitish@nitish-L0Q-15IRX9:~/31359$ hdfs dfs -ls /31359/output
Found 2 items
-rw-r--r--  1 nitish supergroup      0 2026-05-05 18:13 /31359/output/_SUCCESS
-rw-r--r--  1 nitish supergroup    150 2026-05-05 18:13 /31359/output/part-r-00000
nitish@nitish-L0Q-15IRX9:~/31359$ hdfs dfs -cat /31359/output/part-r-00000
.
2
```

```
jar -tf $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar |
grep WordCount
```

```

import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
import org.apache.hadoop.util.GenericOptionsParser;

public class WordCount {

    public static class TokenizerMapper
        extends Mapper<Object, Text, Text, IntWritable>{

        private final static IntWritable one = new IntWritable(1);
        private Text word = new Text();

        public void map(Object key, Text value, Context context
            ) throws IOException, InterruptedException {
            StringTokenizer itr = new StringTokenizer(value.toString());
            while (itr.hasMoreTokens()) {
                word.set(itr.nextToken());
                context.write(word, one);
            }
        }
    }

    public static class IntSumReducer
        extends Reducer<Text,IntWritable,Text,IntWritable> {
        private IntWritable result = new IntWritable();

        public void reduce(Text key, Iterable<IntWritable> values,
            Context context
            ) throws IOException, InterruptedException {
            int sum = 0;
            for (IntWritable val : values) {
                sum += val.get();
            }
        }
    }
}

```

```
    result.set(sum);
    context.write(key, result);
}
}
```

```
public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    String[] otherArgs = new GenericOptionsParser(conf, args).getRemainingArgs();
    if (otherArgs.length < 2) {
        System.err.println("Usage: wordcount <in> [<in>...] <out>");
        System.exit(2);
    }
    Job job = Job.getInstance(conf, "word count");
    job.setJarByClass(WordCount.class);
    job.setMapperClass(TokenizerMapper.class);
    job.setCombinerClass(IntSumReducer.class);
    job.setReducerClass(IntSumReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
    for (int i = 0; i < otherArgs.length - 1; ++i) {
        FileInputFormat.addInputPath(job, new Path(otherArgs[i]));
    }
    FileOutputFormat.setOutputPath(job,
        new Path(otherArgs[otherArgs.length - 1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}
```

```
javac -source 1.8 -target 1.8 -classpath `hadoop classpath` WordCount.java
```